

Home Tutoring Plan v2 — 2-Month Foundation Sprint

Math, Physics & Computer Science | 3 sessions/week × 2 hours = 6 hrs/week | 8 weeks total (24 sessions)

Goal: strong foundations for the degree ahead (not full syllabus coverage — that's a 3.5-year program)

What changed from v1

Timeline cut from 12 months → **2 months (8 weeks, 8 sessions per subject)**.

Real starting points folded in:

CS Session 1 (done): Von Neumann architecture + 1's/2's complement

Physics Session 1 (done): Ohm's Law & Kirchhoff's Laws (circuit basics)

Math: not yet started — begins fresh at Session 1

All three subjects kept **equally weighted** (8 sessions / ~16 hrs each).

With only 8 sessions per subject, each track is trimmed to the **highest-leverage foundation topics** — the ones that unlock everything else in Semester I–II of the actual degree. Deeper/advanced topics (discrete math, OOP, databases, networks, etc.) are deliberately deferred to a Phase 2 plan after this sprint.

Weekly rhythm (unchanged)

Day	Subject
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Day 1 (2h)	Mathematics
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Day 2 (2h)	Physics
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Day 3 (2h)	Computer Science
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MATH — 8 sessions (fresh start)

#	Topic
1	Functions: domain/range, algebra review, intro to limits
2	Limit techniques (factoring, rationalizing) & continuity
3	Derivatives: definition + power/product/quotient/chain rules
4	Applications of derivatives (rates of change, basic optimization)
5	Integration: antiderivatives & basic techniques
6	Vectors: operations, dot product, geometric meaning
7	Matrices: operations, multiplication
8	Solving linear systems (Gaussian elimination) & intro to determinants

Why this order: calculus (1–5) builds the analytical toolkit used everywhere in physics; linear algebra (6–8) is the direct foundation for "Mathematical analysis with linear algebra" (Semester I) and later feeds into graphics, AI, and data structures work in CS.

PHYSICS — 8 sessions (Session 1 already complete)

#	Topic	Status
1	Ohm's Law & Kirchhoff's Laws (circuit basics)	☒ Done
2	Series & parallel circuits, power dissipation	Next
3	Kinematics: 1D motion	
4	Kinematics: 2D motion & projectiles	
5	Newton's Laws & dynamics	
6	Work, energy, power	
7	Momentum & collisions	
8	Capacitors & basic electrostatics (loops back to circuits)	

Why this order: continuing circuits first (2) consolidates what's already started and directly supports "Electrical engineering and electronics." Sessions 3–7 cover core mechanics (the backbone of the general "Physics" course). Session 8 closes the loop back to circuit theory with capacitors/electrostatics.

COMPUTER SCIENCE — 8 sessions (Session 1 already complete)

#	Topic	Status
1	Von Neumann architecture + 1's/2's complement	☒ Done
2	Number systems continued: floating point (IEEE 754 basics), hex/octal conversions	Next
3	Programming fundamentals: variables, data types, if/else	
4	Loops (for/while) & functions	
5	Recursion & problem-solving patterns	
6	Core data structures: arrays & linked lists	
7	Stacks & queues	
8	Intro to algorithms: Big-O notation, basic sorting & searching	

Why this order: finishing number representation (2) closes out the architecture unit cleanly. Sessions 3–5 are "Basics of programming" (Semester I). Sessions 6–8 give an early taste of "Algorithms and data structures" (Semester II) so the student isn't starting from zero when that course begins.

What's deliberately NOT in this 2-month sprint

To hit "strong foundations" in 8 weeks without diluting depth, these are deferred to a **Phase 2 plan** (to be built after this sprint, once we know how the student is tracking):

Discrete mathematics (logic, sets, graph theory, number theory)

Object-oriented programming

Databases & SQL

Operating systems concepts

Computer networks

Software engineering practices

Security/cryptography, AI basics

Waves/optics, electromagnetism (physics)

This isn't a gap — it's sequencing. Everything above builds on what's in this 8-week plan, so nothing is being skipped, just scheduled for after the fundamentals are solid.

Session tracker

Use this to log actual progress vs. plan as we go (update after each session):

Week Math Physics CS

1	☐ M1	☒ P1	☒ C1
2	☐ M2	☐ P2	☐ C2
3	☐ M3	☐ P3	☐ C3
4	☐ M4	☐ P4	☐ C4
5	☐ M5	☐ P5	☐ C5
6	☐ M6	☐ P6	☐ C6
7	☐ M7	☐ P7	☐ C7
8	☐ M8	☐ P8	☐ C8

Next up

Math Session 1 — Functions, domain/range, intro to limits (fresh content, not yet delivered)

Physics Session 2 — Series & parallel circuits, power dissipation

CS Session 2 — Floating point representation (IEEE 754) & hex/octal conversions

Just ask for any of these by name and I'll build the full lesson (explanation + worked examples + practice + take-home questions) the same way as before.